

TRINH HOAI SONG THU

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NUS Computer Science graduate with hands-on experience in full-stack engineering and AI-powered systems, focused on building scalable, intelligent, and user-centric applications.

TECHNICAL SKILLS

Front-End / UI React.js, Next.js, TypeScript, Tailwind CSS, Figma

Back-End Node.js, Express.js, Ruby on Rails, FastAPI, REST APIs

Data & Infrastructure Python, Pandas, R, Spark, Hadoop, MySQL, PostgreSQL, Firebase, MongoDB, Tableau

AI PyTorch, LiteLLM, Hugging Face Transformers

Others Git, Docker, CI/CD tools (GitHub Actions)

EDUCATION

NUS School of Computing, National University of Singapore

Aug 2022 – May 2026

Bachelor of Computing, specialised in Software Engineering, Database Systems, Computer Networks

- Frontend Developer (React) – Issued Apr 2026.
- Claude Code in Action – Issued March 2026.
- SQL (Advanced), HackerRank - Issued Sep 2024.
- Google Data Analytics Program - Issued Aug 2024.
- TikTok Tech Immersion 2024, TikTok - Issued Jul 2024.

WORK EXPERIENCE

Software Engineer Intern, Amaris.AI, Singapore

Jan 2025 – July 2025

- Implemented a full-stack, multi-agent AI threat-hunting platform, integrating Splunk, IntelOwl, and VirusTotal via REST APIs to automate end-to-end threat analysis workflows.
- Designed interactive frontend dashboards enabling real-time monitoring and filtering of security events.
- Built backend services to orchestrate multi-agent AI pipelines, reducing manual threat investigation time by 60%.
- Evaluated SIEM detection rules and AI model defenses, identifying 20+ vulnerabilities across 50+ simulated attack scenarios and achieving 95% detection accuracy.
- Presented the first-phase prototype to clients, demonstrating core platform capabilities and workflow automation.

PROJECTS

Full-stack developer, PeerPrep - Technical Interview Practice Platform | ([github](#))

Sep 2025 – Dec 2025

- Built a real-time collaborative technical interview platform using Next.js, TypeScript, Express.js, MongoDB and WebSocket-based communication.
- Designed a multi-panel UI, enabling seamless navigation during mock interviews and efficient workflow.
- Implemented a matching algorithm pairing users based on skill level and availability with 98% matching accuracy.

Full-stack developer, askWave – University Discussion Platform | ([github](#))

Dec 2023 – Jan 2024

- Built full-stack forum platform for university students using TypeScript, React.js, Ruby on Rails, and PostgreSQL, with core features such as user authentication, discussion threads, and real-time interactions.
- Architected backend APIs and database schemas to support up to 1,000 concurrent users, ensuring reliable performance, efficient queries, and scalable request handling.

Full-stack developer, genius – Student Productivity Web App | ([github](#))

May 2023 – Jul 2023

- Developed responsive web applications using HTML, CSS, JavaScript, Node.js, and MySQL, featuring automatic calendar generation and exam countdowns, reducing students' planning time by 45%.

HACKATHONS

Full-stack developer, phishnchips – BrainHack 2025 Finalist | ([github](#))

Jun 2025

- Finalist among 4,300 students, leading a 3-member team to develop a React Native + Firebase mobile app for multimedia scam detection and reporting.

Data analyst, NUS-RightShip Hackathon

Jan 2025

- Processed, analyzed, and visualized 100K+ environmental records using Python and pandas, uncovering key CO₂ emission trends and optimization opportunities.

EXTRACURRICULAR ACTIVITIES

Volunteer, SASCO West Coast

Aug 2025 – March 2026

- Led a group of 6 volunteers to organize community activities for elderly, fostering social engagement.